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Meta-Analysis and Subgroups

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Abstract Subgroup analysis is the process of comparing a treatment effect for two or more variants of an intervention—to ask, for example, if an intervention's impact is affected by the setting (school versus community), by the delivery agent (outside facilitator versus regular classroom teacher), by the quality of delivery, or if the long-term effect differs from the short-term effect. While large-scale studies often employ subgroup analyses, these analyses cannot generally be performed for small-scale studies, since these typically include a homogeneous population and only one variant of the intervention. This limitation can be bypassed by using meta-analysis. Meta-analysis allows the researcher to compare the treatment effect in different subgroups, even if these subgroups appear in separate studies. We discuss several statistical issues related to this procedure, including the selection of a statistical model and statistical power for the comparison. To illustrate these points, we use the example of a meta-analysis of obesity prevention.

Keywords Meta-analysis · Research synthesis · Systematic review · Subgroups · Subgroup analysis · Fixed-effect · Fixed-effects · Random-effects

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Meta-Analysis and Subgroups

When an intervention is known to be equally effective for all members of a population, it is appropriate to report the impact of that intervention for the population as a whole. By contrast, if the impact varies from one segment of the population to another—for example, if an intervention reduces the risk of an event by 80 % for younger people but by only 20 % for older people—then the analysis must take account of this difference. This is the goal of subgroup analysis.

The term "subgroups" can refer to groupings based on a characteristic of the subjects. As above, we may ask if the treatment is more effective for younger persons than for older ones. Subgroups may also be defined by a variant of the intervention. For example, we may ask if an intervention is more effective when delivered by an outside facilitator rather than a regular classroom teacher. Or, subgroups may be defined by elements of the outcome. For example, we may ask if the long-term effect is different than the short-term effect.

Subgroup analyses in a single study are, of course, limited to the subgroups that are actually included in the study. As such, they are more common in large-scale, multi-center studies, since these often contain multiple subgroups of persons or several variants of the intervention. However, studies of this scope are relatively rare. The more typical situation is that some studies report the treatment effect for one variant of the intervention while others report the treatment effect for a second variant of the intervention, or that some studies enroll one type of person while others enroll a different type of person. When this happens, the only way to compare subgroups is to work with a set of studies rather than a single study. This is the purview of meta-analysis.

Meta-analysis is the process of synthesizing results from an array of studies. The statistical methods used in meta-analysis are analogous to those used in primary studies, except that the unit of analysis is the study rather than the subject. Where a single study will report the mean effect across all subjects, the meta-analysis will report the (weighted) mean effect across all studies. This approach is easily extended to subgroups. We can compute the mean effect for studies that used one variant of the intervention, and also for studies that used another variant of the intervention. Then, we can compare the two (or more) mean effects.

We have two primary goals in this paper. First, we will discuss the statistical models that underlie the computations in a subgroups analysis. An understanding of these is essential in order to choose the appropriate options in software, and to correctly interpret the results. Second, we will discuss how these models affect the statistics, and the implications for statistical power. To illustrate these points, we use the example of a meta-analysis of obesity prevention.

Interventions to Prevent Morbid Obesity

Obesity has reached epidemic proportions and poses an important threat to public health in much of the developed world. Padwal et al. (2003a, b) conducted a series of meta-analyses to assess the potential role of drugs in addressing this threat, and the example that follows is based on data presented in these papers. For the purposes of this paper, we will focus on the technical aspects of the analysis rather than substantive issues such as the inclusion criteria for the studies. The reader who is interested in the substantive issues should consult the original publications.

Figure 1 shows the results of an analysis to compare the impact of two drugs, Orlistat and Sibutramine.

The outcome in this analysis is the proportion of persons who reduced their weight by at least 5 %, and the effect size is the difference in proportions for the treated vs. control groups. For example, the study by Lindgarde (2000) reported that 54 % of the treated persons met this milestone, as compared with 41 % of the controls. The difference is 13 percentage points with a 95 % confidence interval of 3 to 23 percentage points.

The top section of the plot shows results for 14 studies (with a total of 9,389 persons) that compared Orlistat vs. placebo. The next section of the plot shows results for seven studies (with a total of 1,464 persons) that compared Sibutramine vs. placebo. The bottom shows results for the 21 studies (with a total of 10,853 persons) as a whole.

Immediately below each section, the plot shows the summary effect for that section. The effect size is represented by a square and bounded by a horizontal line that reflects the precision (95 % confidence interval) of the estimate. For Orlistat, the effect is 21 percentage points (18 to 24). For Sibutramine, the effect is 32 percentage points (27 to 37). For the sample as a whole, the effect is 24 percentage points (21 to 28).

The effect size estimates for each subgroup (above) are based on a particular statistical model. Similarly, the comparison of effects (below) will depend on a particular statistical model. Our goal is to discuss what the possible models are, how to select among them, and how to interpret the results in the context of the selected model.

Statistical Models in a Simple Meta-Analysis

Before turning to subgroups, consider a simple meta-analysis, where we want to compute the summary effect for a single collection of studies. The summary effect M is computed as the weighted mean of the study effect sizes,

$$M = \frac{\sum_{i=1}^{k} W_i Y_i}{\sum_{i=1}^{k} W_i} \tag{1}$$

and the variance of the summary effect is computed as

$$V_M = \frac{1}{\sum_{i=1}^{k} W_i} \tag{2}$$

where W_i is the weight assigned to study i, and Y_i is the effect size for study i. In Eq. 1, the relative weight assigned to each study influences the value of the summary effect size. In Eq. 2, the absolute weight assigned to each study influences the precision of the summary effect size. As such, the weights play an integral role in the computations.

There are various statistical models employed in meta-analysis, and the selection of a model determines how we assign a weight to each study. Researchers typically talk about two statistical models for a simple meta-analysis: fixed-effect and random-effects. For the purposes of this paper, we need to discuss three models: fixed-effect (singular), fixed-effects (plural), and random-effects.

The Fixed-Effect (Singular) Model

Suppose we sample k studies from a population of studies, but the population is limited to studies that are (essentially) identical to each other, and as such are assumed to share a common effect size. Our intent is to estimate this common effect size, which we will generalize to the population of all essentially identical studies. This case is depicted in Fig. 2. The true effect size for each study is represented by a circle, and in keeping with the assumption of a common effect size, all the circles line up above the common value (θ) of 50.0. The observed value for each study is represented by a square, and the squares differ from θ because of sampling error.

The amount of sampling error is suggested by the normal curve for each study. Each curve is centered on the corresponding study's true effect size, and the curve's width reflects the standard error in estimating that effect size. The weight assigned to each study is

$$W_i = \frac{1}{V_i} \tag{3}$$

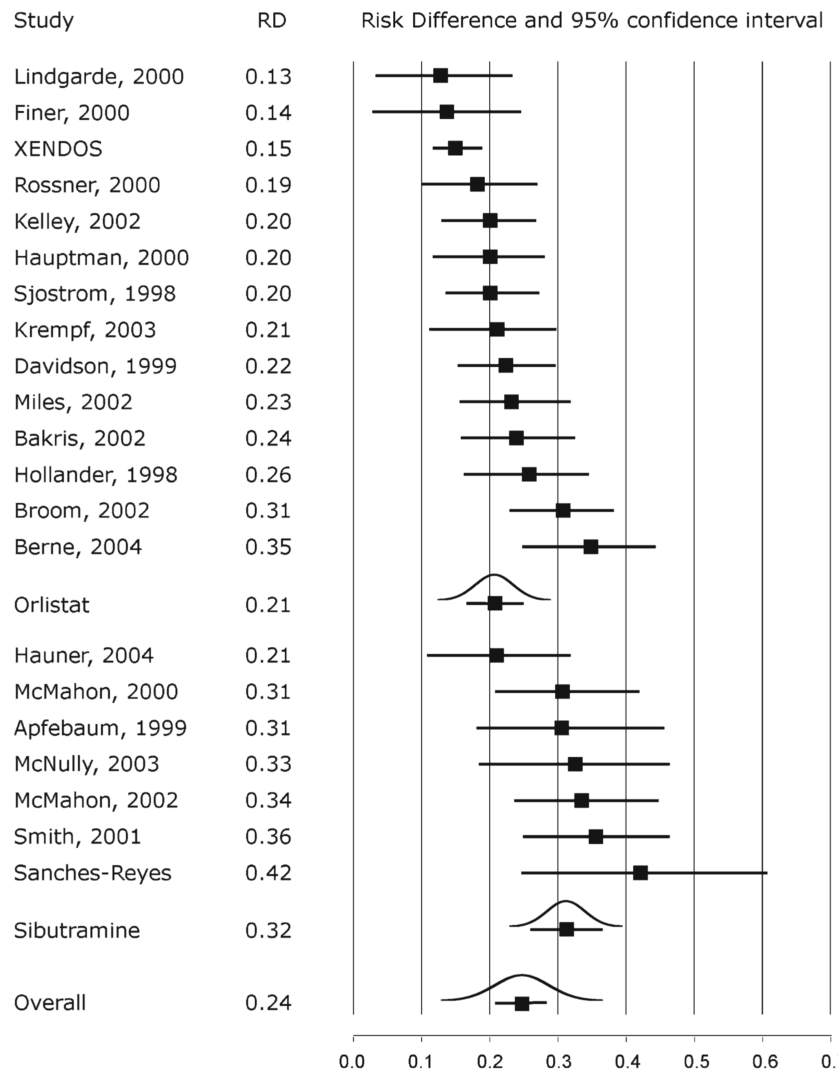


Fig. 1 Impact of two drugs in effecting weight loss (based on Padwal et al. 2003a, b)

where V_i is the variance of the mean for study i. The variance tends to be smaller for large studies (here, study 2), and so these tend to be assigned more weight in the analysis. (Note that the word variance sometimes refers to the variance of subject scores and sometimes to the variance of the mean.)

In this model, while there is sampling at the study level, there is no variance at the study level. Since all studies in the

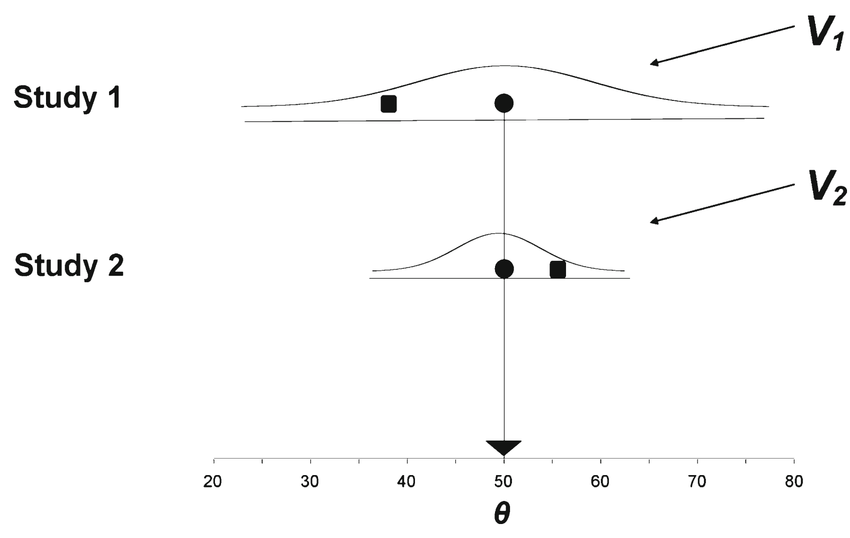


Fig. 2 Schematic of the fixed-effect (singular) model

population (as we have defined the population) share a common effect size, there is no sampling error introduced by the fact that we happened to include one study rather than another. The label "fixed-effect" can be interpreted as "common-effect," and "effect" is singular since there is only one effect size.

The Fixed-Effects (Plural) Model

Suppose we include k studies in the analysis, and the true effect size differs from study to study. Our intent is to report the mean effect (μ) for these k studies, but we have no interest in generalizing beyond them. This case is depicted in Fig. 3. The true effect size for each study is represented by a circle, with values of 45 and 55. The observed effect for each study is represented by a square, and differs from that study's true effect because of sampling error.

The amount of sampling error is again suggested by the normal curve for each study, and the weight assigned to each study is again

$$W_i = \frac{1}{V_i} \tag{4}$$

where V_i is the variance of the mean for study i.

While the variance (and the formula for the weight) are the same here as for the fixed-effect (singular) model, the reason is different. In the case of the fixed-effect model, there was sampling at the study level but no variance. For the fixed-effects model, there is variance at the study level but no sampling. Since we have defined the population as being this particular set of studies, there is no sampling error introduced by the fact that we included these studies and not others. The model is called "fixed-effects" because the effects are "fixed" in the sense that they are "set." "Effects" is plural since the true effect size may vary from study to study.

The Random-Effects Model

Suppose we sample k studies from a population of studies where the true effect size (θ) differs from study to study. Our intent is to report the mean effect (μ) for the population of studies. This case is depicted in Fig. 4. The true effect size for each study is represented by a circle, with values of 45 and 55. These circles differ from the true mean (50) because of sampling error between studies. The observed effect for each study is represented by a square, and differs from that study's true effect because of sampling error within studies.

In this case there are two sources of sampling error. The amount of sampling error within studies (the fact that the subjects within each study are sampled from a population of subjects) is again suggested by the normal curve for each study. The amount of sampling error between studies (the fact that the effects underlying the studies in the analysis are sampled from a population of scenarios) is suggested by the normal curve about μ near the bottom of the schematic.

The weight assigned to each study is

$$W_i = \frac{1}{V_i + T^2} \tag{5}$$

where V_i and T^2 are the estimated within-study variance and between-studies variance, respectively. The name "random-effects" reflects the assumption that the effects underlying the studies have been randomly sampled from a population of scenarios. The word "effects" is plural since the true effects are assumed to vary.

Fig. 3 Schematic of the fixed-effects (plural) model

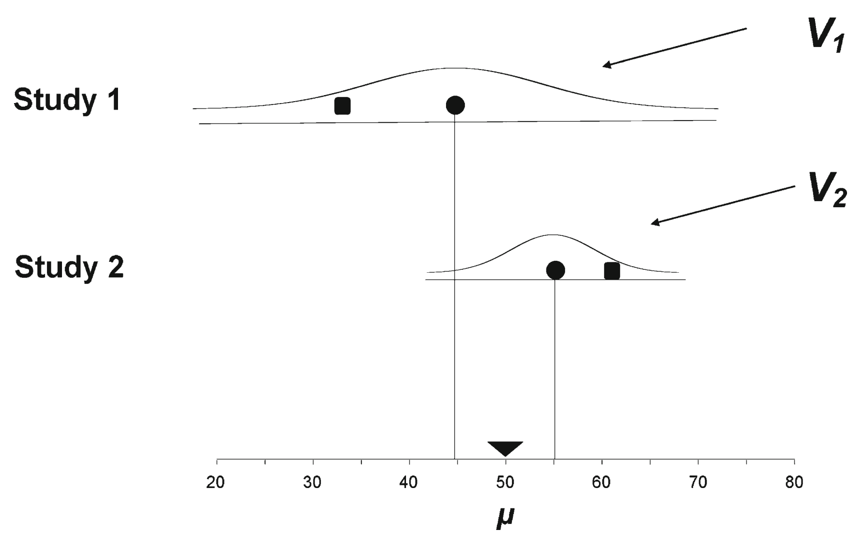
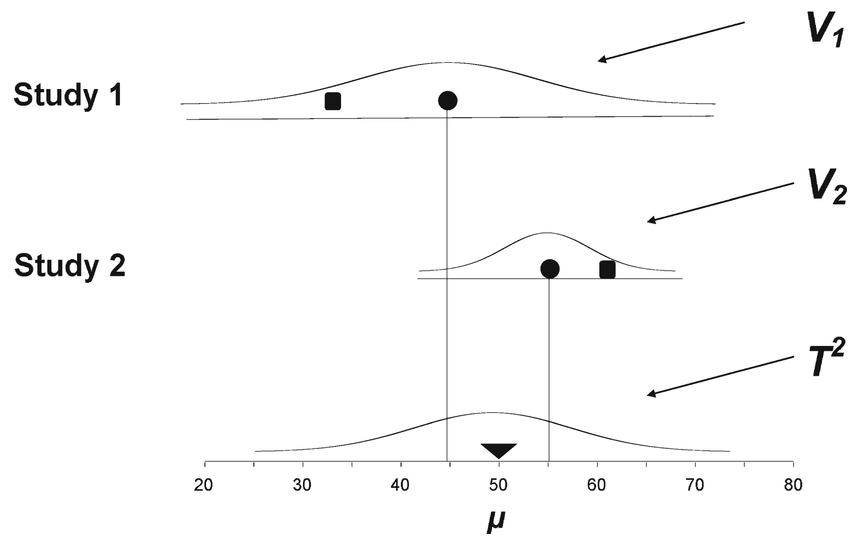


Fig. 4 Schematic of the random-effects model



Selecting a Model in the Simple Case

Which model should we use to compute a summary effect?

Fixed-effect (singular) This model is appropriate if we assume that all studies in the analysis share a common effect size. It follows that we are estimating a single parameter and that the analyses need to take account of only one source of sampling error. For example, suppose that a pharmaceutical company enrolls 1,000 persons for a trial. It then randomly assigns each person to one of ten cohorts, so that it can work with 100 people at a time. The same researchers will be responsible for all cohorts and there are no seasonal nor practice effects. The studies are identical (for all intents and purposes), and the fixed-effect model should be applied. However, this kind of synthesis is relatively rare.

Fixed-effects (plural) This model is not usually appropriate because (by definition) the results cannot be generalized beyond the studies in the analysis. However, if the goal really is to report solely on the studies at hand (for example, as part of a pilot study or to meet a regulatory requirement), then this could be the appropriate model. Additionally, this model is sometimes pressed into service when one would prefer to use the random-effects model but cannot obtain a sufficiently precise estimate of between-studies variance to do so comfortably. In this case, the researcher might prefer to report the mean for the studies at hand (where at least the precision of the estimate is known) rather than report the mean for the larger population (with a possibly spurious estimate of precision).

Random-effects In the vast majority of meta-analyses, and certainly in analyses where the studies have been performed by researchers working independently of each other, the assumption of a common effect size (required for the fixed-effect model) is not tenable. Additionally, we want to generalize beyond the studies at hand, and therefore the fixed-effects model is not appropriate. The random-effects model does not require the assumption of a common effect size, and it does allow us to generalize beyond the studies at hand. Therefore, in most cases, it provides the best match to the sampling frame and to our goals.

The differences among these models are outlined in Table 1.

The selection of one model or the other (random-effects vs. either of the "fixed" models) will impact the estimate of the summary effect. Small studies will tend to have more impact under random-effects than under either of the fixed models, while large studies will tend to have more impact under the fixed models (for example, Borenstein et al. 2010).

The selection of a model (random-effects vs. either of the "fixed" models) will also impact on the precision of the summary effect. For simplicity, assume that we are working with single-group studies with a normally distributed variable, and that all studies have the same sample size and the same standard deviation s. Under the fixed models, the variance of the summary effect for k studies is estimated as

$$V_M = \frac{s^2}{n} \tag{6}$$

where n is the cumulative sample size across all studies. By contrast, under the random-effects model, the variance of the summary effect for k studies is estimated as

$$V_M = \frac{s^2}{n} + \frac{T^2}{k} \tag{7}$$

where T^2 is the estimated variance of effects across studies, and k is the number of studies. It follows that the variance of the summary effect will always be as wide or wider under the random-effects model as compared with either of the fixed models.

Table 1 Sources of variance in a simple meta-analysis

	Within-study variance	Between-studies variance	Total variance	Relevant
Fixed-effect	Yes	No (no variance)	V_{Y_i}	Limited
Fixed-effects	Yes	No (no sampling)	V_{Y_i}	Limited
Random-effects	Yes	Yes	$V_{Y_i} + T^2$	Yes

Subgroup Analysis

With this as background we can proceed to the issue of subgroups. There are two distinct parts to a subgroups analysis: (a) computing the effect within each subgroup and (b) comparing the summary effects across subgroups. We will address each separately.

Within Subgroups

To compute the summary effect within each subgroup, we need to select either the fixed-effect (singular) model, the fixed-effects (plural) model, or the random-effects model. The definition of each model, and the criterion for selecting one or another, is the same as it was for the simple meta-analysis.

Fixed-effect (singular) In a simple analysis (one with a single set of studies), the fixed-effect model is appropriate when logic dictates that the studies share a common effect size. Similarly, in a subgroups analysis, the fixed-effect model can be employed within subgroups when logic dictates that these studies share a common effect size. For the simple case, we presented the example of a pharmaceutical company that performed a series of essentially identical studies, and we can extend that example to apply to subgroups. If the same company wanted to run ten essentially identical studies that compared drug A vs. control, and ten essentially identical studies that compared drug B vs. control, then the fixed-effect model would be appropriate within each subgroup. However, as was true in the simple meta-analysis, this scenario is relatively rare.

Fixed-effects (plural) In a simple meta-analysis, the fixed-effects model can be employed when the studies at hand represent the full population of interest. Similarly, in a subgroups analysis, the fixed-effects model can be employed within subgroups when studies within each subgroup represent the full population of interest for that subgroup. For the simple analysis, this model may be appropriate in specific cases and is sometimes employed as a fallback position when there are problems in using the random-effects model. The same logic applies within subgroups.

Random-effects In a simple analysis, the random-effects model is appropriate when we expect (or allow) that the true effect size varies from study to study. Similarly, in a subgroups analysis, the random-effects model should be employed within subgroups when logic dictates that the effect size varies within subgroups. While we usually expect that the effect sizes within a subgroup will be more homogeneous than effect sizes across subgroups, there is generally no reason to expect that the studies within a subgroup will have exactly the same (true) effect size. Therefore, as for a simple analysis, this model is often the best fit.

Which model should we select? In the obesity example, there is no reason to expect that the effect size is identical across studies within subgroups, and so the fixed-effect model does not apply. In addition, we do want to generalize beyond the studies at hand, and so the fixed-effects model does not apply. Rather, the random-effects model is the best fit for the data. Additionally, there are enough studies to allow for a reasonably accurate estimate of between-studies variance and so there are no technical barriers to applying the random-effects model. Therefore, this is the model that was employed to compute the effect size and confidence intervals within subgroups as presented in Fig. 1 and excerpted in Fig. 5 and Table 2. Computational details are presented in the Electronic supplementary material.

Between Subgroups

After we have computed the effect size within subgroups, we need to compare these effect sizes. For this purpose, we need to compute the standard error of the difference, SE_{Diff}. This will enable us to report a confidence interval and significance test for the difference in effects.

The formula for SE_{Diff} depends on the statistical model that tells us how the subgroups were sampled. As before, we can discuss the fixed-effect (singular), fixed-effects (plural), and random-effects models. When we were working with a single set of studies or within subgroups (Table 1), we were concerned with variance within and between studies. Now, we are concerned with variance within and between subgroups, but the logic is the same.

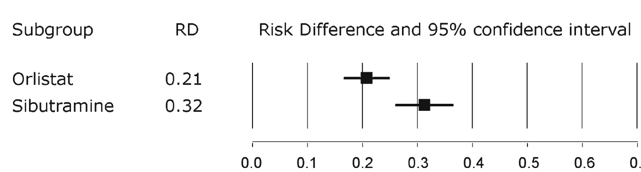


Fig. 5 Summary effect and confidence interval for two drugs

Table 2 Summary effect and confidence interval for two drugs

Drug	Risk difference	Variance	Standard error	Lower limit	Upper limit
Orlistat	0.213	0.000228	0.015	0.183	0.242
Sibutramine	0.321	0.000731	0.027	0.268	0.373

Fixed-effect (singular) When we were working within subgroups, the fixed-effect model was appropriate when we assumed that all studies within the subgroup shared a common effect size. When we are working between subgroups, this model would require that all subgroups share a common effect size. This assumption is not tenable—indeed, the whole point of the analysis is to compare the various summary effects. Therefore, this model is not an option.

Fixed-effects (plural) When we were working within subgroups, the fixed-effects model was appropriate when our interest was limited to the studies at hand. Similarly, when we are working between subgroups, this model is appropriate when our interest is limited to the subgroups at hand. However, whereas this was not generally a plausible assumption within subgroups, it is usually a perfectly reasonable assumption across subgroups. Indeed, it may be the only plausible model for most analyses in medical research, prevention research, and some other fields.

For example, suppose that we want to compare the impact of an intervention for subgroups based on gender. The question of interest is "Is the effect size different for males vs. females?" Any researcher who wanted to ask the same question would need to use the same two subgroups, and therefore, there is no sampling error introduced by the fact that we selected these subgroups rather than others. To put this in context, if we have an infinite number of studies within each subgroup, so that we knew the effect for each subgroup with no error, then we would also know the difference in effects with no error.

Random-effects When we were working within subgroups, the random-effects model is appropriate when the studies are sampled from a population of studies. Similarly, when we are working between subgroups, this model is appropriate when the subgroups are sampled from a population of subgroups. For example, suppose universities around the country have each run k experiments to test the impact of an intervention, and we want to know if the effect varies by university. Each experiment is a study, and each university is a subgroup. The question of interest is "Does the effect size vary by university?" Researcher X performs a subgroup analysis to compare the effect across universities and randomly selects universities A and B. Researcher Y performs a similar analysis and randomly selects universities C and D. The fact that the subgroups are sampled introduces another source of sampling error. To put this in context, suppose that each of the universities in these analyses included an infinite number of studies. We would know the difference between A and B exactly and would also know the difference between C and D exactly, but we would not know the difference among all possible pairs of universities exactly. Indeed, the results for Researchers X and Y would not match.

In examples such as this, the random-effects model for subgroups may be viewed as part of a multilevel model (e.g., Raudenbush and Bryk 2002), with subjects nested within studies and studies nested within universities. In such models, we are often more interested in accounting for differences between universities than in estimating differences between particular universities.

Computing the standard error of the difference Table 3 shows how the variance of the difference depends on both the within-subgroups model and the between-subgroups model.

The variance of the difference always incorporates the within-subgroups variances, V_{M_A} and V_{M_B}, as represented by the three rows in Table 3. Since V_{M_A} and V_{M_B} are computed within subgroups, the within-subgroups model always affects these values.

The variance of the difference also depends on the between-subgroups model as represented by the columns in Table 3. If we are using the fixed-effects model between subgroups, then variance of the difference is estimated as

$$V_{\text{Diff}} = V_{M_A} + V_{M_B}. \tag{8}$$

By contrast, if we are using the random-effects model between subgroups, then the variance of the difference is estimated as

$$V_{\text{Diff}} = V_{M_A} + V_{M_B} + \frac{T_G^2}{m}. \tag{9}$$

The difference between Eqs. 8 and 9 is the addition of T_G^2/m in the latter. The statistic T_G^2 is the estimate of between-subgroups variance (where the subscript G represents subgroups), and m is the number of subgroups. The expression T_G^2/m represents the variance added to the error term by the fact subgroups are sampled rather than fixed.

Note: These formulas are based on an analysis of single-group studies where the effect size (or point-estimate) is the group mean. Additionally, these formulas require that all the studies share the same sample size and the same standard deviation. We chose this example because here, $V_{M_A} = V_{M_B} = s^2/n$, which allows us to

Table 3 Sources of variance for the difference between subgroups

Within each of m subgroups, n subjects are divided equally among k studies

Within-subgroups model		Between-subgroups (difference) model: Fixed-effects	Between-subgroups (difference) model: Random-effects
Fixed-effect	$V_{M_A} = V_{M_B} = s^2/n$	$V_{\text{Diff}} = V_{M_A} + V_{M_B}$	$V_{\text{Diff}} = V_{M_A} + V_{M_B} + T_G^2/m$
Fixed-effects	$V_{M_A} = V_{M_B} = s^2/n$		
Random-effects	$V_{M_A} = V_{M_B} = s^2/n + T^2/k$		

highlight the parallel between s^2/n, T^2/k, and T_G^2/m. However, we would not apply these formulas in practice. Rather, we would use formulas based on the actual within-study variance, which is based on the metric being used in the analysis and also other particulars of the studies.

Which model should we select between subgroups? In the obesity example, the question was framed as "What is the difference in effect between Orlistat and Sibutramine?" Since the question is about these two drugs, the subgroups are "fixed" rather than sampled, and the fixed-effects model is appropriate.

Working with the values for V_{M_A} and V_{M_B} from Table 2 and applying Eq. 8, we can estimate the variance of the difference as

$$V_{\text{Diff}} = 0.000228 + 0.000731 = 0.000959 \tag{10}$$

and

$$\text{SE}_{\text{Diff}} = \sqrt{0.000959} = 0.031 \tag{11}$$

Then, given the difference between subgroups of

$$\text{RD} = 0.320 - 0.213 = 0.108,$$

the 95 % confidence limits for the difference are

$$\text{LL}_{\text{Diff}} = 0.108 - 1.96 \times 0.031 = 0.047$$
$$\text{UL}_{\text{Diff}} = 0.108 + 1.96 \times 0.031 = 0.168$$

A test of the null hypothesis that the effect size is identical for Orlistat and Sibutramine is given by

$$Z = \frac{0.108}{0.031} = 3.478,$$

with a two-tailed p value of

$$p(Z) = 0.000505.$$

Computational details are provided in the Electronic supplementary material.

In Context

In the obesity example, we selected the random-effects model within subgroups and the fixed-effects model between subgroups. Note that the selections correspond to the sampling process.

- Precision within subgroups depends on the precision within studies and also on the number of studies. Even if we knew the effect size for each study with no error, we would still not know the mean effect size for the subgroup with no error. This corresponds to the random-effects model within subgroups.
- Precision of the difference depends on the precision within subgroups but not on the number of subgroups. If we knew the effect size for each subgroup with no error, we would also know the difference between them with no error. This corresponds to the fixed-effects model between subgroups.

Statistical Power

The statistical models influence the standard error of the difference. If our goal is to report the magnitude of the difference, the model affects the confidence interval for this estimate. If our goal is to test the null hypothesis that the two drugs are equally effective, the model affects the power of the test.

In their papers on statistical power for comparing subgroups, Hedges and Pigott (2001, 2004) note that some researchers believe that when we use meta-analysis to compare the effect across subgroups, statistical power is uniformly high. This leads to the assumption that if the difference is not statistically significant, the effect in the two subgroups must be similar. In fact, though, while power for comparing subgroups may be high, that is not always the case. Rather, power for these kinds of analyses can be very low for several reasons.

One reason is that the effect size for a subgroups analysis is based on the differential effects (the difference between effect sizes), and therefore (in most cases) is smaller than the effect size for a main effect.

A second reason is the fact that precision for estimating the effect size, a key component in statistical significance (and power), may be poor. In any given analysis, there will be one, two, or three distinct sources of error, and we need to consider the impact of each separately. The following points are evident in Table 3.

- In all cases, the precision of the difference is affected by s^2/n, the within-study variance and the cumulative number of subjects. However, this source of error dissipates as the cumulative number of subjects increases.
- If we are using a random-effects model within subgroups, then the precision of the difference depends also on T^2/k, the between-study variance and the cumulative number of studies. Even if the sample size is in the tens of thousands, precision will be limited to a value determined by T^2/k, and power may be very low. This applies to the row labeled "Random-effects."
- If we are using a random-effects model between subgroups, then the precision of the difference depends also on T_G^2/m, the between-subgroups variance and the number of subgroups. Even if there are hundreds of studies within each subgroup, the precision will be limited to a value determined by T_G^2/m, and power may be very low. This applies to the column labeled "Random-effects."

Nomenclature

As noted earlier, researchers often think of two possible models (fixed vs. random). With these two options within subgroups and the same two options between subgroups, there are four possible combinations, and these are typically labeled "Fixed," "Mixed," and "Random."

However, these labels can be confusing. For example, the label "Fixed" suggests that the same model is being used at both levels of the analysis, when in fact the researcher might be using the fixed-effect (singular) model at one level and fixed-effects (plural) at the other. The label "Mixed-effects" may refer to fixed-effect within and random-effects between, or random-effects within and fixed-effects between (among other options).

Therefore, rather than using these labels, it would be better to refer to each option by explicitly naming the models as in "Random-effects within, Fixed-effects between."

Statistical and Methodological Notes

Assumptions Required for the Various Models

Above, we noted that the assumptions underlying the fixed-effect model are rarely met, and suggested that the random-effects model is usually a better fit for the data in a simple meta-analysis or within studies. Others have argued that while the assumptions of a fixed-effect model are rarely met, the same can be said for a random-effects model. For example, it may not be reasonable to assume that the effects underlying the studies performed are a random sample of all realistic effect sizes. As a separate issue, the estimate of τ^2 may be incorrect (for example, Bonett 2008, 2009). These issues are currently being debated in the literature with regard to simple meta-analysis, and these debates are relevant to subgroup analysis as well.

Subgroup Analysis is Observational

In a single study where persons are randomly assigned to either treatment or control, any difference between groups can usually be attributed to the treatment. The same holds true when we use a meta-analysis to yield a combined estimate of that same effect. In other words, the estimate in the meta-analysis is protected by the same randomization that covers each of the individual studies.

However, this protection does not extend to the subgroup analyses since the moderators are characteristics of the studies, and are not assigned at random. In our example, Sibutramine proved more effective than Orlistat. While this may be because the former is actually more effective, it is also possible the persons enrolled in the Sibutramine studies were more predisposed to benefit from drug treatment and that Sibutramine's superior performance was (at least partly) due to this confound. Therefore, the difference between subgroups is observational and subject to the caveats that normally attend to observational studies.

Working with More than Two Subgroups

In primary studies, we generally work with t tests to compare effects in two subgroups, and with F tests (analysis of variance) to compare effects across more than two subgroups. The two are based on the same statistics, and will yield identical results when applied to the two-group case.

The same applies to subgroups analysis in meta-analysis. We have presented statistics for the two-group case because these are relatively transparent and allowed us to focus on the models. However, all the computations outlined in this paper can be extended to the case of three or more subgroups. In that case, rather than working with the ratio of a mean difference to its standard error, we work with the ratio of variance across subgroups relative to variance within subgroups. Details are presented in Borenstein et al. (2009), Hedges and Olkin (1985), Lipsey and Wilson (2001), and Cooper et al. (2009).

Summary

The obesity analysis serves as an example of how meta-analysis can be used to compare subgroups. Since the individual studies reported the impact of Orlistat or Sibutramine but not both, they could not provide information on the relative impact of the two. By contrast, meta-analysis makes this comparison possible. In addition to making the analysis possible, meta-analysis often yields a richer, more informative picture than would be possible with two subgroups in a single study. Usually (as in this example), the meta-analysis yields a more precise estimate of summary effect in each subgroup (and of the difference in effects) than any single study. Additionally, the meta-analysis provides context, telling us (for example) that the true effects within each subgroup fall within a certain range, and providing a sense of the overlap in effects between the subgroups.

This is not to imply that meta-analyses are preferable to single studies in all respects. In particular, when single studies incorporate multiple subgroups, they may allow for head-to-head comparisons, which is an important advantage. However, when several studies provide these direct comparisons, then these comparisons may be included in a meta-analysis as well.

All of this, of course, requires that the appropriate statistical models are employed in the analysis. When the model's assumptions are met, the statistics are valid. Otherwise, they are not. In this paper, we have discussed the various options and proposed a framework for choosing among them.

We outlined the impact of statistical models on power. The fact that one model yields better power than another should never be a factor in model selection. Different models address different questions, and the only relevant issue is to select the model that matches the question at hand. Still, it is important to understand how the different models are related to statistical power. This allows the researcher to appreciate the fact that power may be low even in an analysis with many subjects and studies, and also to design meta-analyses in ways that are conducive to higher power and better precision.

Finally, we explained that subgroup analyses are observational by nature, and that this is true even when all studies within the subgroups employ random allocation.

The approach outlined in this paper has many possible applications in the field of prevention science. These include asking if an intervention's impact is affected by the setting, by the delivery agent, or by the quality of delivery, among many others.

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